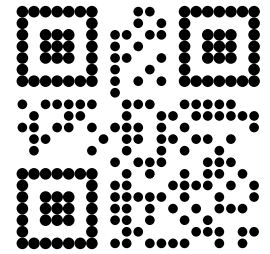


Cassidy Soloff

Website: csoloff.com | email: cassidy@csoloff.com



Education

PhD in Atmospheric Science

University of Arizona, Tucson, AZ

Minor: Remote Sensing; GPA: 4.00

Anticipated Dec 2026

MS in Atmospheric Science

University of Arizona, Tucson, AZ

GPA: 4.00

Aug 2022 – May 2024

BA Physics and Astronomy

Wesleyan University, Middletown, CT

Minor: Environmental Studies; GPA: 3.87

High Honors Thesis: <https://doi.org/10.14418/wes01.1.2557>

Aug 2018 – May 2022

Professional Experience

Graduate Research Assistant

Advisor: Dr. Armin Sorooshian

University of Arizona, Tucson, AZ

May 2022 – Present

- NASA's Aerosol Cloud meTeorology Interactions oVer the western ATlantic Experiment (ACTIVATE)
 - Researched the spatial evolution of aerosol and gas properties of the U.S. East Coast using flights transiting from Virginia to Bermuda (Publication: <https://doi.org/10.5194/acp-24-10385-2024>)
 - Predicting Cloud Condensation Nuclei (CCN) concentration from aerosol chemistry using κ -Köhler theory and compared with direct CCN measurements to assess understanding of aerosol microphysics and instrumentation (Publication: <https://doi.org/10.1021/acsestair.5c00151>)
- NASA's Arctic Radiation-Cloud-Aerosol-Surface-Interaction Experiment (ARCSIX)
 - Conducted lab study using Differential Aerosol Sizing and Hygroscopicity Spectrometer Probe (DASH-SP) to measure deliquescence and refractive index of different particle species
 - Collected data with DASH-SP instrument during ARCSIX mission in Northern Greenland during summer 2024, flying 28 combined hours on the NASA P-3 Orion aircraft
 - Diagnosed and resolved DASH-SP issues in the field
 - Developed scripts to process raw data from DASH-SP

NASA Student Airborne Research Program Intern

Advisors: Dr. Roya Bahreini, Andreas Beyersdorf; Mentor: Eva-Lou Edwards

• Researched the local radiative forcing from the 2018 Kīlauea flank eruption

• Flew 10 combined hours on NASA DC-8 research aircraft to monitor the aerosol instruments

Jun 2021 – Aug 2021

Undergraduate Research Assistant

Advisor: Dr. Seth Redfield

Wesleyan University, Middletown, CT

Jan 2020 – May 2022

- Modeled exoplanet atmosphere spectra and developed an exoplanet transit survey for hot, bright stars
- Worked as an Undergraduate Research Fellow in the Summer of 2020, assisting Dr. Seth Redfield as a co-investigator for his James Webb Space Telescope proposal to make spectral observations
- Conducted the first exoplanet transit observations with the Van Vleck Observatory's 24-inch PlaneWave telescope to derive the photometric sensitivity and precision of the new equipment
- Analyzed exoplanet transit curves to derive stellar and planetary properties

Telescope Operator

Feb 2019 – Jan 2020

Wesleyan University, Middletown, CT

- Operated the 24-inch Perkin telescope and software to produce science images of stars and galaxies in specified wavelength bands, providing data for Wesleyan researchers

Teaching Experience

Astronomy Public Outreach

Feb 2019 – May 2022

Wesleyan University, Middletown, CT

- Led presentations for the department's Space Nights and assisted in public outreach events for kids

Course Assistant: Introduction to Astronomy

Sep 2021 – Dec 2021

Wesleyan University, Middletown, CT

- Assisted students during office hours and graded homework/exams

Volunteering

Treasurer

Aug 2023 – May 2024

University of Arizona, Tucson, AZ

- Managed finances for outreach and social events for Hydrology and Atmospheric Science Student Association

President of Astronomy Club

Sep 2021 – May 2022

Wesleyan University, Middletown, CT

- Organized club activities and presented weekly on current topics in Astronomy

Professional Development

Carbon Methods Workshop

Apr 2023

The Nature Conservancy in Arizona

- Performed measurements of tree and soil carbon along the lower San Pedro River

Peer-Reviewed Publications

8. Taylor, P. C., Sorooshian, A., Ueyama, R., Schmidt, S., Abboud, I., Allison, Q., Barry, K., Becker, S., Bender, H. A., Bennett, J. R., Blair, J. B., Bohn, N., Boisvert, L., Brown, M. D., Brintjes, R., Bucholtz, A., Buzanowicz, M., Cairns, B., Calì Quaglia, F., Chemyakin, E., Chen, B., Chen, G., Chen, H., Chen, Y. W., Cheng, Z., China, S., Chirica, D., Choi, Y., Colarco, P., Collister, B., Crosbie, E., Cross, M. J., Daniels, J., DeMott, P., DiGangi, J. P., di Sarra, A. G., Diskin, G. S., Dolinar, E. K., Edwards, E. L., Ephraim, S., Evangeliou, N., Foskinis, R., Gallo, F., Gao, L., Gómez-Amo, J. L., Gonzalez, D., Groot Zwaafink, C., Gupta, P., Heckman, I., Hendrickson, M., Hilario, M. R. A., Hirata, K., Hofton, M., Holen, A. L., Im, U., Khan, A. L., Kahn, R., Korolev, A. V., Kreidenweis, S., Krumpen, T., Kurtz, N., Lait, L.,

- Lamkin, B., Landy, J., Lata, N. N., Lawson, P., LeBlanc, S., Leavor, S., Li, J., Markus, T., Maring, H., Massling, A. H., Mavis, C., McGinnity, F., Meyer, K., Mojica, G., Moore, R. H., Morris, P., Muscari, G., Nataraja, V., Nehrir, A. R., Nenes, A., Nowottnick, E. P., Ottaviani, M., Parker, C., Patnaude, R., Perez, M., Perkins, R. J., Peterson, C., Petty, A., Phothisane, S., Polashenski, C., Pratt, K. A., Preisler, K. M., Prytherch, J., Rabine, D., Redemann, J., Ryoo, J. M., Schlosser, J. S., Selimovic, V., Shook, M. A., Silverman, M., Skov, H., Smirnov, A., **Soloff, C.**, Solomon, A., Stamnes, S., Takeishi, A., Thompson, D. R., Thornhill, K. L., Tilling, R., Tjernström, M., Tosco, M., Valdelomar, P. C., Van Gilst, D., Wang, J., Wang, Z., Wasilewski, A., Wendisch, M., Wilder, B., Winstead, E. L., Wu, A., Xian, P., Zamora, L. M., Zhang, J., Zhang, L., Zhang, L., Ziemba, L., & Zuidema, P. (2026). The Arctic Radiation-Cloud-Aerosol-Surface Interaction Experiment (ARCSIX) airborne campaign dataset. *Earth System Science Data Discussions*, 2026, 1-77. <https://doi.org/10.5194/essd-2026-241>
7. **Soloff, C.**, Crosbie, E. C., Diskin, G. S., Gao, L., Kirschler, S., Lenhardt, E. D., Moore, R. H., Redemann, J., Shook, M. A., Tang, S., Thornhill, K. L., Voigt, C., Wang, H., Winstead, E. L., Ziemba, L. D., & Sorooshian, A. (2025). Cloud Condensation Nuclei Behavior and Closure Assessment for the Northwest Atlantic Ocean. *Environmental Science & Technology Air*. <https://doi.org/10.1021/acsestair.5c00151>
 6. Lenhardt, E., Gao, L., Hostetler, C. A., Ferrare, R. A., Burton, S. P., Moore, R. H., Ziemba, L. D., Crosbie, E., Sorooshian, A., **Soloff, C.**, & Redemann, J. (2025). Aerosol effective radius governs the relationship between cloud condensation nuclei (CCN) concentration and aerosol backscatter. *Atmospheric Chemistry and Physics*, 25(20), 13747-13768. <https://doi.org/10.5194/acp-25-13747-2025>
 5. Sorooshian, A., Siu, L. W., Butler, K., Brunke, M. A., Cairns, B., Chellappan, S., Chen, J., Choi, Y., Crosbie, E. C., Cutler, L., DiGangi, J. P., Diskin, G. S., Ferrare, R. A., Hair, J. W., Hostetler, C. A., Kirschler, S., Kleb, M. M., Li, X.-Y., Liu, H., McComiskey, A., Namdari, S., Painemal, D., Schlosser, J. S., Shingler, T., Shook, M. A., Silva, S., Sinclair, K., Jr, W. L. S., Soloff, C., Stamnes, S., Tang, S., Thornhill, K. L., Tornow, F., Tselioudis, G., Van Dienenhoven, B., Voigt, C., Vömel, H., Wang, H., Winstead, E. L., Xu, Y., Zeng, X., Zhang, B., Ziemba, L., & Zuidema, P. (2025). The NASA ACTIVATE Mission. *Bulletin of the American Meteorological Society*, BAMS-D-24-0136.0131. <https://doi.org/10.1175/BAMS-D-24-0136.1>
 4. Namdari, S., Ajayi, T., Choi, Y., Crosbie, E. C., DiGangi, J. P., Diskin, G. S., Kirschler, S., Liu, H., Nowak, J. B., Shook, M. A., Soloff, C., Thornhill, K. L., Voigt, C., Winstead, E. L., Zhang, B., Ziemba, L. D., & Sorooshian, A. (2024). A comprehensive analysis of new particle formation across the northwest Atlantic: Analysis of ACTIVATE airborne data. *Atmospheric Environment*, 338, 120831. <https://doi.org/10.1016/j.atmosenv.2024.120831>
 3. **Soloff, C.**, Ajayi, T., Choi, Y., Crosbie, E. C., DiGangi, J. P., Diskin, G. S., Fenn, M. A., Ferrare, R. A., Gallo, F., Hair, J. W., Hilario, M. R. A., Kirschler, S., Moore, R. H., Shingler, T. J., Shook, M. A., Thornhill, K. L., Voigt, C., Winstead, E. L., Ziemba, L. D., & Sorooshian, A. (2024). Bridging gas and aerosol properties between the northeastern US and Bermuda: analysis of eight transit flights. *Atmospheric Chemistry and Physics*, 24(18), 10385-10408. <https://doi.org/10.5194/acp-24-10385-2024>
 2. Rzos, J. L., Fernández-Valenzuela, E., Ortiz, J. L., Rommel, F. L., Sicardy, B., Morales, N., Santos-Sanz, P., Leiva, R., Vara-Lubiano, M., Morales, R., Kretlow, M., Alvarez-Candal, A., Holler, B. J., Duffard, R., Gómez-Limón, J. M., Desmars, J., Souami, D., Assafin, M., Benedetti-Rossi, G., Braga-Ribas, F., Camargo, J. I. B., Colas, F., Lecacheux, J., Gomes-Júnior, A. R., Vieira-Martins, R., Pereira, C. L., Morgado, B., Kilic, Y., Redfield, S., Soloff, C., McGregor, K., Green, K., Midavaine, T., Schreurs, O., Lecossois, M., Boninsegna, R., Ida, M., Le Cam, P., Isobe, K., Watanabe, H., Yuasa, S., Watanabe, H., & Kidd, S. (2024). A study of centaur (54598) Bienor from multiple stellar occultations and rotational light curves. *Astronomy & Astrophysics*, 689, A82. <https://doi.org/10.1051/0004-6361/202450833>

1. Ajayi, T., Choi, Y., Crosbie, E. C., DiGangi, J. P., Diskin, G. S., Fenn, M. A., Ferrare, R. A., Hair, J. W., Hilario, M. R. A., Hostetler, C. A., Kirschler, S., Moore, R. H., Shingler, T. J., Shook, M. A., Soloff, C., Thornhill, K. L., Voigt, C., Winstead, E. L., Ziemba, L. D., & Sorooshian, A. (2024). Vertical variability of aerosol properties and trace gases over a remote marine region: a case study over Bermuda. *Atmospheric Chemistry and Physics*, 24(16), 9197-9218. <https://doi.org/10.5194/acp-24-9197-2024>

Presentations

- (Oral)** Soloff, Cassidy, et al. "Characterization of Dry Aerosol Real Refractive Index in the Arctic Surrounding Spring–Summer Sea Ice Melt." ARCSIX Science Team Meeting, Hampton, VA, Apr 2026.
- (Poster)** Soloff, Cassidy, et al. "Hygroscopic and optical properties of cloud droplet residual particles in the Arctic atmosphere: Vertically-resolved measurements from the ARCSIX mission." ARCSIX Science Team Meeting, Hampton, VA, Apr 2026.
- (Poster)** Soloff, Cassidy, et al. "Seasonal and Chemical Influences on Dry Refractive Index and Hygroscopic Growth Factor." ARCSIX Science Team Meeting, Boulder, CO, May 2025.
- (Poster)** Soloff, Cassidy, et al. "CCN Closure for the ACTIVATE campaign." El Dia del Agua y la Atmosfera, Tucson, AZ, March 2025.
- (Oral)** Soloff, Cassidy, et al. "CCN Closure for the ACTIVATE campaign." American Geophysical Union, Fall Meeting, Washington D.C., December 2024.
- (Oral)** Soloff, Cassidy, et al. "Bridging Gas and Aerosol Properties between Northeast U.S. and Bermuda: Analysis of Eight Transit Flights." El Dia del Agua y la Atmosfera, Tucson, AZ, March 2024.
- (Poster)** Soloff, Cassidy, et al. "Characterization of Trace Gases and Particles Across the Northwest Atlantic: Analysis of Transit Flights during the NASA ACTIVATE Mission." American Geophysical Union, Fall Meeting, San Francisco, CA, December 2023.
- (Oral)** Soloff, Cassidy, et al. "Bridging Gas and Aerosol Properties between Northeast U.S. and Bermuda: Analysis of Eight Transit Flights." ACTIVATE Science Team Meeting, Tucson, AZ, November 2023.
- (Poster)** Soloff, Cassidy, et al. "Characterization of Trace Gases and Particles Across the Northwest Atlantic: Analysis of Transit Flights during the NASA ACTIVATE Mission." El Dia del Agua y la Atmosfera, Tucson, AZ, March 2023.
- (Poster)** Soloff, Cassidy, et al. "Radiative Forcing Analysis of the 2018 Kīlauea Flank Eruption." American Geophysical Union, Fall Meeting, Chicago, IL, December 2022.
- (Poster)** Soloff, Cassidy, and Seth Redfield. "Unseen Worlds: A Search for Exoplanet Transits of Bright Early-Type Stars." American Astronomical Society, Pasadena, CA, June 2022.
- (Panel Discussion)** Redfield, S., Bennett, K., Soloff, C., McGregor, K., STEAMfest, Bridgeport, CT, Nov 2021

Scholarships, Fellowships & Honors

- | | |
|---|----------|
| Galileo Circle Student Scholarship | Apr 2026 |
| <i>"The Galileo Circle awards scholarships to undergraduate and graduate students who demonstrate exceptional potential in the physical, mathematical, environmental, cognitive, or life sciences."</i> | |
| Galileo Circle Student Scholarship | Apr 2025 |
| Sol Resnick Scholarship | Aug 2024 |
| Scholarship established in memory of Dr. Sol D. Resnick awarded to deserving graduate students in the Department of Hydrology and Atmospheric Sciences | |
| Roots for Resilience Fellowship | Aug 2024 |

"The Roots for Resilience Program provides training and support to select graduate students on open, reproducible science and computational infrastructure tools to enhance research focused on resiliency in the environment."

Galileo Circle Student Scholarship Apr 2024

NASA Group Achievement Award Mar 2023

"2022 Agency Honor Awards for an outstanding group accomplishment that has contributed substantially to NASA's mission"

El Día del Agua y La Atmosfera Best Poster Mar 2023

Awarded by Pima County Environmental Quality

Graduate College Fellowship Mar 2023

The Graduate College allocates Graduate College Fellowship funding to the academic college deans as block grants to recruit and retain top international or domestic graduate students.

Littell Prize May 2022

"The gift of Franklin Bowers Littell, Class of 1891, for excellence in one or more advanced courses in astronomy."

NASA Undergraduate Travel Grant Dec 2021

Awarded by NASA Connecticut Space Grant Consortium (CTSGC) for travel to the American Astronomical Society

NASA Undergraduate Research Grant May 2021

Awarded by NASA CTSGC to support research for the exoplanet transit survey

Wesleyan Undergraduate Research Fellowship May 2020

Funding to research at Wesleyan University with Dr. Seth Redfield in the summer of 2020

Reviewer Roles

Journal of Geophysical Research – Atmosphere Dec 2026

Atmospheric Environment Sep 2025

Atmospheric Chemistry and Physics Sep 2025

Qualifications & Skills

Skills: Proficient in Python, ArcGIS Pro; Basic in R, C

Instrumentation: DASH-SP, PlaneWave 24-inch telescope

Strengths: Data analysis, machine learning, leadership, communication, teamwork, organization